# Image Analysis

An Open-Source Summary

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# Contents

# Part I.
# Classic Image Analysis

## 1. Introduction

The key take away is that vision is subjective to the human being and themselves. It is the results of continuous evolution and survival of the fittest. We are more sensitive to light and certain colors, to motion, … And this fact will be used for efficient image representation and analysis.

We can think of the classic illusion where our brain will interpret certain lengths of an arrow longer than others. Context also does matter, even in a blurry picture our brain will train to make up for what he sees.

### 1.1. The basics

We first need light to have vision. Light is influenced by object and create perception. ADD image slide 51. The light source bounce on the object and some scattered ray will get trough the 2D Image plane, which will be casted on the sensor array thanks to the lens system.

#### 1.1.1. Light

We will look at it from a physical optic (not geometrical (constant speed and straight path) nor quantum). Light is an Electro-magnetic wave which we assume it to be planar. aka $\frac{|E|}{|H|} = \eta$ in simple term, light is perpendicular. This is a **self-sustaining process** that is characterized by:

1. $\lambda$: wavelength
2. Direction
3. $E$: Amplitude
4. $\varphi$: phase
5. Direction of polarisation

Violet: low frequency ($380nm$); Red: high frequency ($760nm$). Remember $\lambda = \frac{c}{f}$.

#### 1.1.2. Perception

We don't perceive light equally. Humans have 3 cones that are not evenly spread apart. But the human vision is trained to compensate for it so everything feels similar in intensity to us. Some animals can have more cones.

#### 1.1.3. Interaction with matter

**Table 1:** The four types of interaction

| Phenomenon | Example |
|---|---|
| Absorption | Blue water,green leaf |
| Scattering | blue sky, red sunset |
| Reflection | colored in k |
| Refraction | dispersion by a prism |

**1.1.3.1. Absorption** Dissipation of $\lambda$ specific for the medium. Resonance in molecules.

**1.1.3.2. Scattering** When light (radiation/particles) deviates from a straight trajectory by local non-uniformities. Type depends on the relative size of particles and wavelengths:

1. **Rayleigh**: small ($\lambda$ dependent)
2. **Mie**: comparable (weakly $\lambda$ dependent)
3. **Non-selective**: large ($\lambda$ **in**dependent)

We ignored transmission and diffraction.

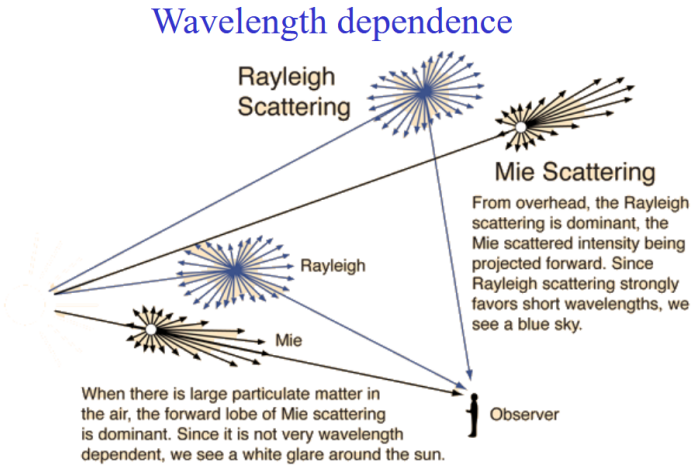



**Figure 1:** Wavelength dependence for scattering

We need scatter to get the bluesky (Rayleigh with Tyndall effect) or it would be dark. Coloured cloud from volcanic eruption = Mie (particle all over the air). Grey cloud is due to non-selective.Less scatter for lower frequency light (infrared).

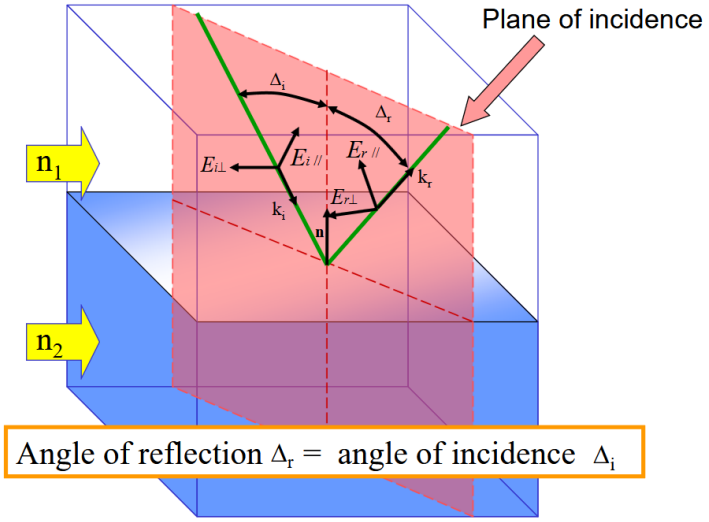



**Figure 2:** Mirror reflection

**1.1.3.3. Reflection** With reflection, the notion of polarization is important. The polarization effect creates the fact that a reflected wave on certain surfaces can be reflected with all but one direction of the electric field. This effect is more prominent on **non-metallic** surfaces!

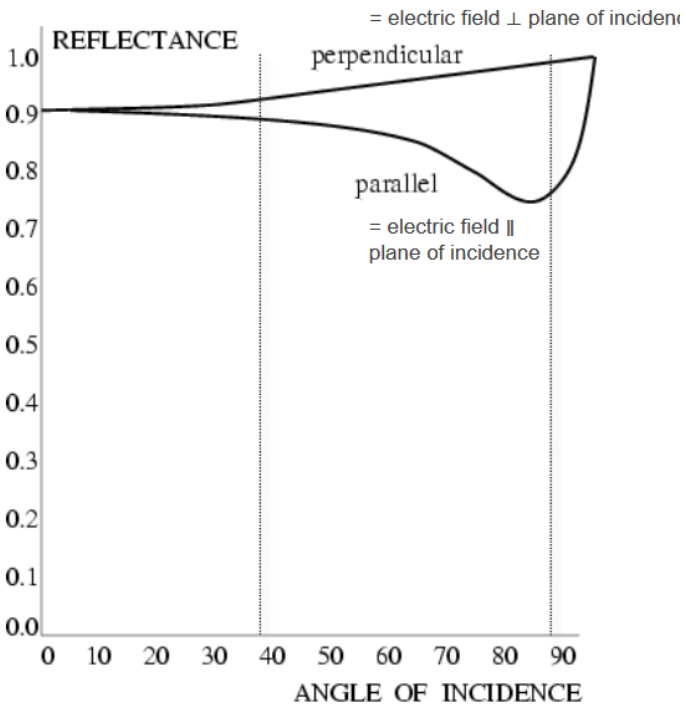



**Figure 3:** Non-metallic reflection - strong reflectors

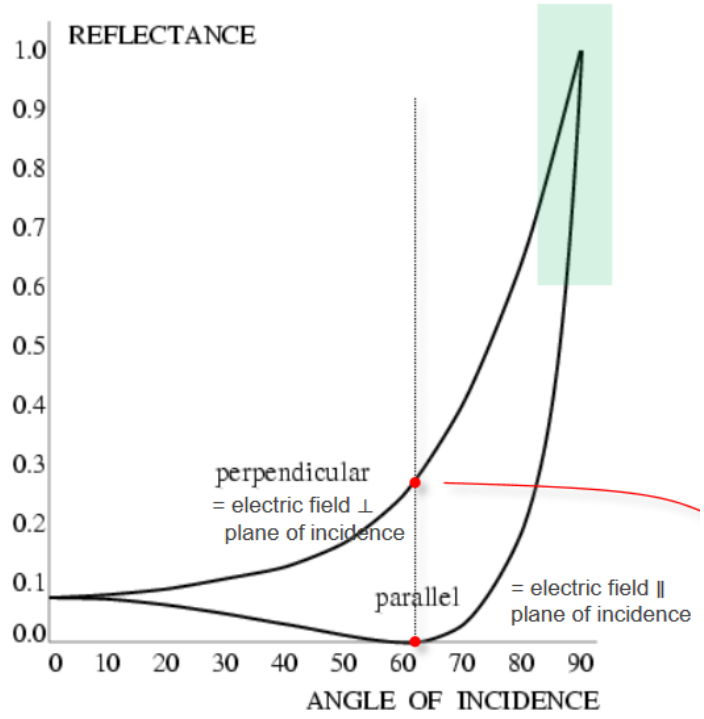



**Figure 4:** Metallic reflection - Red dot: Brewster angle (polarizer) - Blue zone: Grazing angles

Rough surfaces will lead to **diffuse** reflection as the reflection angles will be all over the place. On the other hand, smooth surfaces give **specular** reflection. Both **diffuse** and **specular** can also happen all together.

Reflectance varies through $\lambda$ and different side of vegetation can typically give various reflection[1]

**1.1.3.4. Refraction** Typically, refraction which is the bending of the light and dispersion which is a frequency dependency.

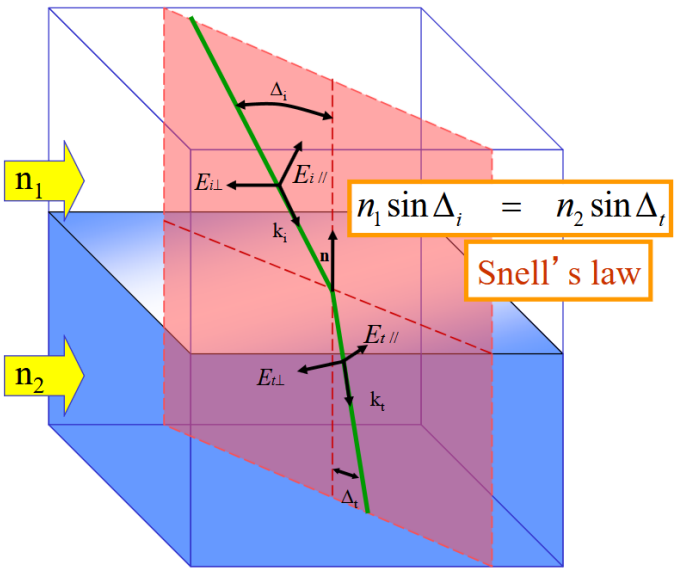



**Figure 5:** Refraction - governed by Snell's law property of the medium **and** material

[1] This fact can be used for spectral reflectance of vegetation since it forms a signature

## 2. Acquisition of images

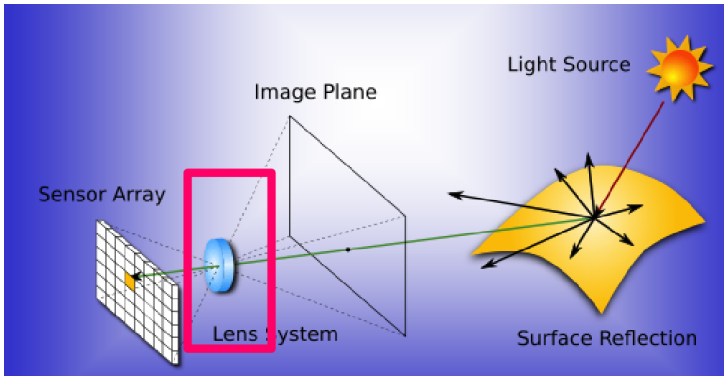



**Figure 6:** Image acquisition model

Controlling the lights is essential for acquisition as it is the main vector to acquire images. Severals techniques exist

| Type | Description | Examples |
|---|---|---|
| Back-lighting | Light source behind object (better with a diffuser plate) | High contrast silhouette. **Binary vision**, inspection |
| Directional | Tilt lights to generate sharp shadow of specular reflection (rough surfaces) | Detection of cracks |
| Diffuse | Uniform lighting, all directions contribute to the diffusion reflection but specular spike due to spike | Detection of cracks |
| Polarized: Contrast diffuse and specular | diffuse makes light non-polarized while specular keeps polarized. Put polarizer/analyzer orthogonal, better for glare (high dynamic range) issue. | Detection of cracks |
| Polarized: Contrast dielectrics and metal | Dielectric at Brewster angle can be used as a polarizer which is not possible for metallic surface (see previous chapter). | So use a polarizer to suppress specular reflection from dielectric **OR** distinguish between specular and dielectric surfaces |
| Coloured | Highlight region of similar colour. Use laser (monochromatic light), differentiation between specular and diffuse. Comparing colours. Spectral sensitivity of a sensor. | |
| Structured | Spatially or temporally modulated light pattern. Projection on a 3D object will distort the projection pattern | 3D reconstruction |
| Stroboscopic | High intensity light flash. Eliminate motion blur. | For high speed inspection |

Note about the dar and bright field. Mostly relevant for the Directional/Diffuse lighting. *Dark field*, when the camera is placed outside the specular reflection area of the normal area, only specular reflection (different orientation from normal)

will show up. On the other hand, *Bright field*, we place camera inside this specular reflection light area and the non-normal area will appear dark.

## 2.1. Image formation - Lens

We use the pinhole model in this class:

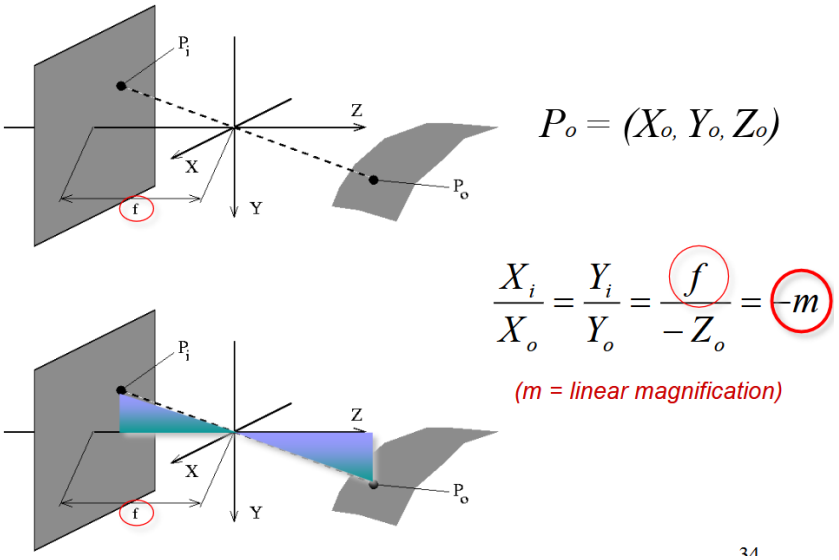



**Figure 7:** Pinhole model

From this we can use the thin-lens equation which assumes:

- Spherical lens surfaces
- Incoming light $\pm \parallel$ to axis
- Thickness « radii
- Same refractive index on both sides

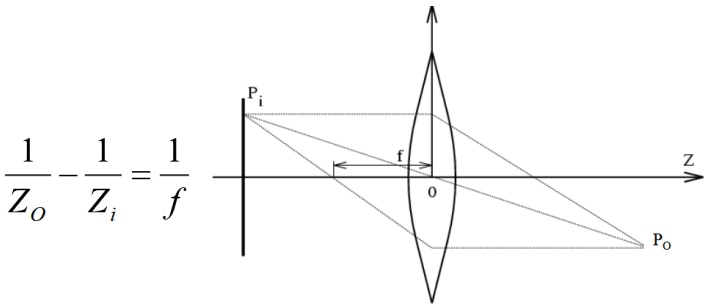
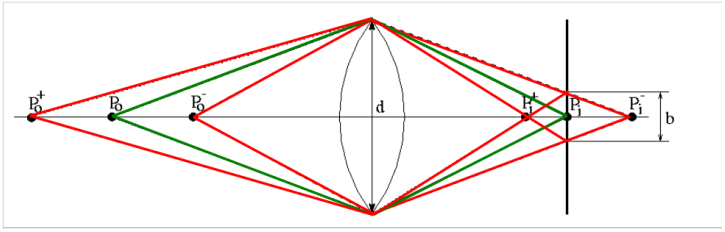



**Figure 8:** Thin-lens equation




$$\Delta Z_0^- = Z_0 - Z_0^- = \frac{Z_0(Z_0 - f)}{Z_0 + f\ d/b - f}$$

**Figure 9:** Depth of field of the thin lens

There is one focal point. Strike a balance between incoming light (d) and large depth-of-field. Smaller depth-of-field is present in microscopes for example.

## 2.2. Sensor array

### 2.2.1. CCD vs CMOS

We have 2 types of sensors:

- CCD: Charge-coupled device
  - Old school
  - Charges must hop from tile to tile
  - Cover the total pixel (high fill rate)
  - Expensive
  - Blooming (due to the charge hoping)
  - High power consumption
- CMOS: integrated
  - Cheap, Can put other component on the chip (integration)
  - Read line by line (like RAM = random access)
  - Smaller
  - Low power
  - Same sensor as CCD but each has their own amplifier = more noise + not covering the full space = lower fill rate / sensitivity

They both works in 2 steps:

1. Photon to electron (charge) conversion
2. Electron (charge) to voltage conversion

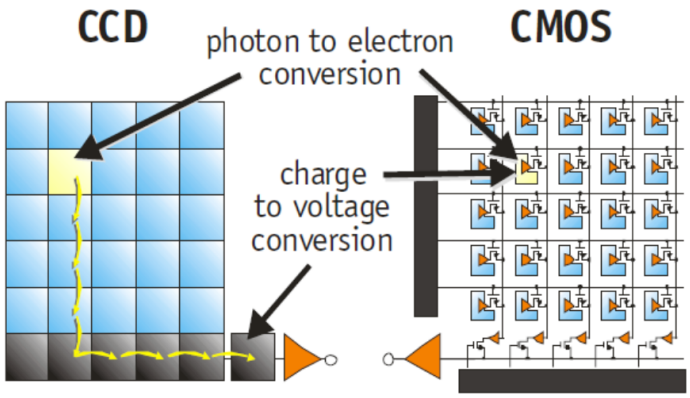



**Figure 10:** CCD vs CMOS

### 2.2.2. Colour cameras

**2.2.2.1. 1. Prism** The idea is to split the light using *dichroic prism* to split it in 3 colors. This requires really good precision but allows for good color separation.

**2.2.2.2. 2. Filter mosaic** Back to the *bayer pattern* introduced earlier. Good for tricking the human eyes. Each pixel will have a filter in front to follow the bayer pattern. This is good but can create some aliasing especially when an edge is right on a pixel array causing some extra inexistant colors (aliasing).

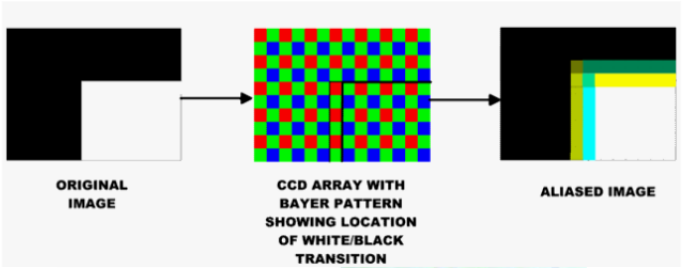



**Figure 11:** Aliasing with bayer pattern

We loose actual resolution since know we need 2x2 pixel array to get 1 colored pixel ! We have to stack even more density but need to capture more lights –> Use microlenses.

**2.2.2.3. 3. Filter wheel** Here instead of having 3 filters like in a filter mosaic, we have a full wheel with many filters. Behind the wheel, there is a static sensor (**only static scenes**).

#### 2.2.2.4. Summary

**Table 3:** Summary of the 3 coloured camera

| Type | Prism | Mosaic | Wheel |
|---|---|---|---|
| # Sensors | 3 | 1 | 1 |

| Type | Prism | Mosaic | Wheel |
|---|---|---|---|
| Resolution | High | Medium | Good |
| Cost | High | Low | Medium |
| Framerate | High | High | Low |
| Artefacts | Low | Aliasing | Aliasing (just motion of wheel) |
| Bands (color) | 3 | 3 | 3 or + |
| Application | High-end cameras | Low-end cameras | Scientific applications |

**2.2.2.5. Improvements** One issue is the Aliasing problem for object in motion. Since we read line by line in CMOS camera, for fast turning object (turning at $\geqslant f_s/2$) we have aliasing. To solve this, we need a global shutter instead of reading line by line.

## 2.3. Perspective Projection - Image Plane

Here, we revisit the pinhole model where now, we first project everything on an object plane and then this will get through the lens. The center of the lens is the center of the projection, we have an virtual image plane which is also called **perspective projection** (we flattened out everything from the outside on it):

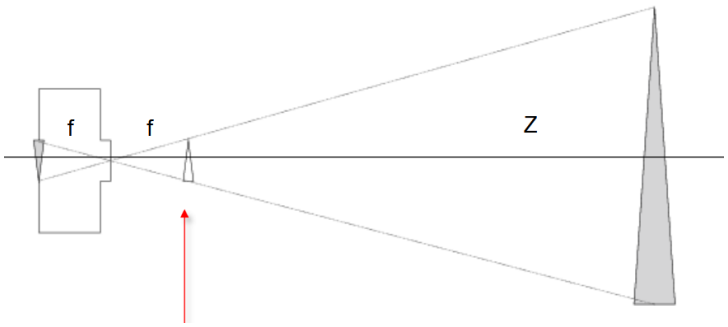



**Figure 12:** Camera projection model

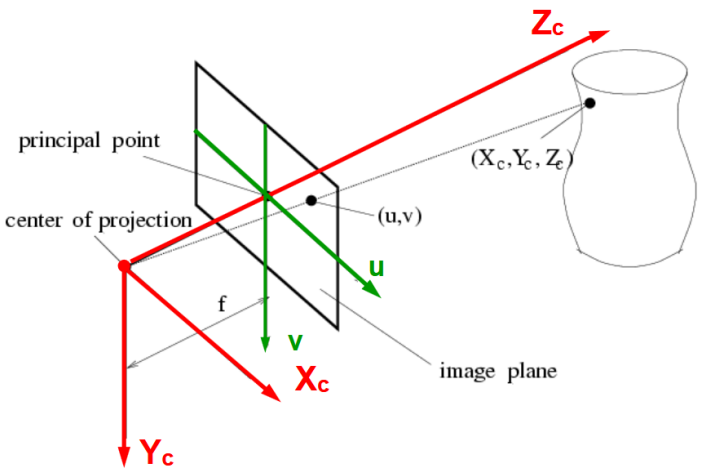



**Figure 13:** 3D view of the camera projection model

The origin is the center of the projection ($X_c$ row-pixel, $Y_c$ column-pixel space), the $Z_c$ axis correspond with the optical axis.

If we look at the $u, v$ space we have the following relationship, which comes from the "similar triangles property" where you look once from above (for X case) and one from the side (for Y case).

$$u = f\frac{X_c}{Z_c} \qquad v = f\frac{Y_c}{Z_c}$$

We can then say that if $Z$ is constant, we can write $x = kX_c$ and $y = kY_c$ where $k = f/Z$. Orthographic projection + a scaling. This forms a good approximation if objects are small compared to their distance from the camera.

### 2.3.1. Projections limitations

**Figure 14:** Pseudo-orthographic vs Perspective projection

The perspective projection model is incomplete. What about world coordinate ? What if we want our u,v coordinates as a row and column numbers (like in a matrix of pixels).

**This part of the class that does not form exam material but helps with understanding where does those matrices come from.**

**2.3.1.1. The (u,v) projection to pixel** We can see a matrix of pixel instead of the (u,v) projection. We just need 3 informations: 1. Center of matrix $x_0$ and $y_o$, 2. Width of array $k_x$ and 3. Height of array $k_y$ giving us:

$$\begin{cases} x & = k_x u + x_0 + sv \\ y & = ky_v + y_0 \end{cases}$$

Note: the $s$ is for skew, we have the aspect ratio $k_y/k_x$[2]

**2.3.1.2. The global world coordinate problem** Those equations are based on the previous $u = fX/Z$ and $v = fY/Z$. Here $\langle , \rangle$ represents the dot product $\cdot$ between 2 vectors.

[2] I thought it was the opposite…

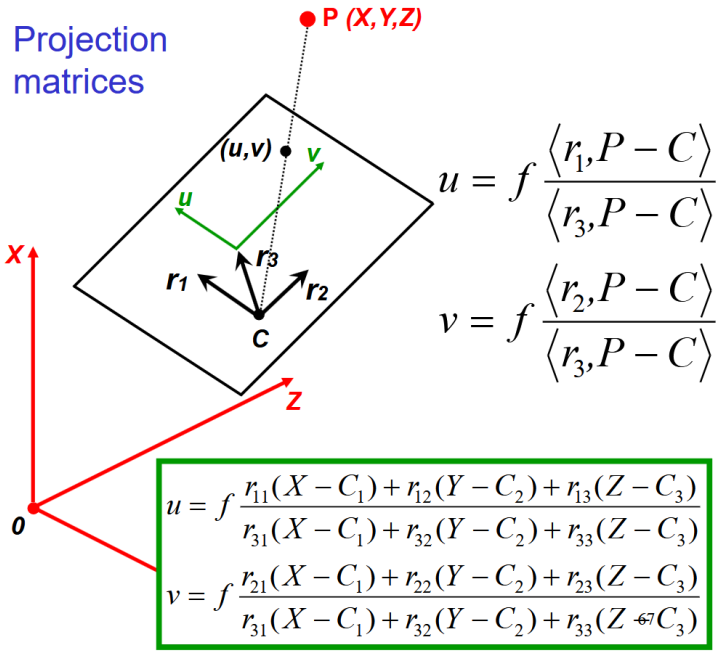



**Figure 15:** Projection matrices

The $P - c$ represents the dotted vector and $C$ being the camera position in a global coordinate system (the camera is no longer stuck to the 0,0 point). We then use the dot product between on of its axis and the vector P-C to obtain the value in this specific vector dimension (a bit like the X before). Repeat it for Y and Z.

**2.3.1.3. Homogeneous coordinates** Capturing a 2D image requires to navigate a 3D world (and so on for 3D and 4D). This is important in computer vision where we have to deal with projection where we **have to** divide a point by $z$ !

$$\tau \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow_{\text{capturing/projection}} \begin{pmatrix} x/z \\ y/z \end{pmatrix}$$

**2.3.1.4. Putting it all together** We can rewrite the previous big dot/inner product using the homogeneous coordinate (removes division) as:

$$\tau \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} fr_{11} & fr_{12} & fr_{13} \\ fr_{21} & fr_{22} & fr_{23} \\ r_{31} & r_{12} & r_{13} \end{pmatrix} \begin{pmatrix} X - C_1 \\ Y - C_2 \\ Z - C_3 \end{pmatrix}$$

we can then apply the pixel coordinate system on this equation:

$$\tau \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} k_x & s & x_0 \\ 0 & k_y & y_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} fr_{11} & fr_{12} & fr_{13} \\ fr_{21} & fr_{22} & fr_{23} \\ r_{31} & r_{12} & r_{13} \end{pmatrix} \begin{pmatrix} X - C_1 \\ Y - C_2 \\ Z - C_3 \end{pmatrix}$$

**Back to important material for exam.**

we can further simplify with

$$\tau \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \overbrace{\begin{pmatrix} k_x & s & x_0 \\ 0 & k_y & y_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix}}^{K \text{ Calibration matrix}} \overbrace{\begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{12} & r_{13} \end{pmatrix}}^{R \text{ Camera rotation}} \begin{pmatrix} X - C_1 \\ Y - C_2 \\ Z - C_3 \end{pmatrix}$$

We then define the following vectors:

$$p = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \qquad P = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \qquad \tilde{P} = \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \tag{1}$$

Which then yields the simplified version:

$$\rho p = KR^\top (P - C) \qquad \text{some non-zero } \rho \in \mathbb{R}$$

$$\text{or, } \rho p = K(R^\top | - R^\top C)\tilde{P}$$

$$\text{or, or, } \rho p = (M|t)\tilde{P} \qquad \text{with rank } M = 3$$

## 2.4. Deviations from the lens model

We assume:

1. All rays from a point are focused onto 1 image point
2. All image points in a single plane
3. Magnification is constant

If those are not respected, we have **aberrations**! Aberrations acan be **geometrical** (small for paraxial rays) or **chromatic** (refractive index function of $\lambda$ - Snell's law)

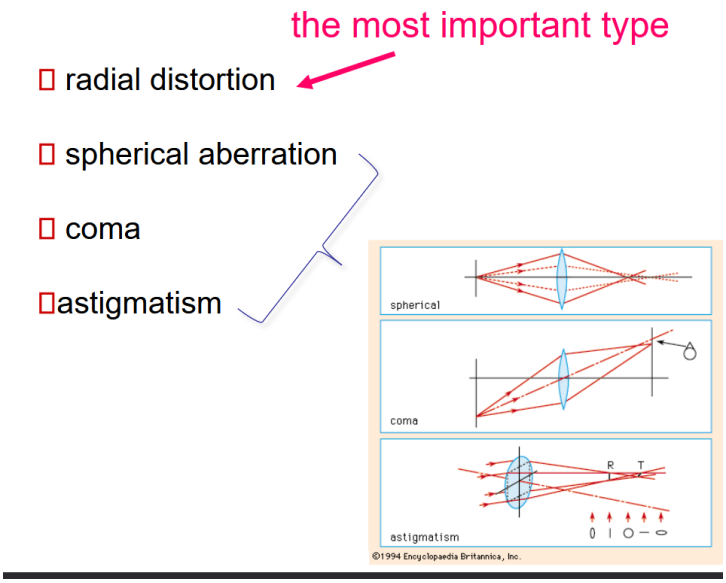



**Figure 16:** Geometrical aberrations

They all cause issues on the focus point of the image plane. Spherical appears when the bending of light is not uniform throughout the lens, typically outer ray converges sooner (small focal length) then inner one. They do not converge all together. Coma is a bit like spherical but this time with a tiled ray. Astigmatism is problems between the *tangential* and *sagittal* ($90°$ rotation from tangential) focus point which are not equal.

### 2.4.1. Radial distortion

By far the most important type of distortion which causes “fish-eye” look or pincushion. It creates various *magnification* for different angles of inclination. The pixel either sinks towards the center or slides outside along the line connection the center and the original pixel.


**Figure 17:** Radial Distortion

We can model this sliding distance $d$ with the following equation and thus can be corrected. Some algorithm looks at human made structure (straight lines) to establish the effect.

$$d = (1 + \kappa_1 r^2 + \kappa_2 r^4 + ...)$$

#### 2.4.2. Chromatic aberration

Rays of different $\lambda$ will focus in different planes. This creates various colours especially along sharp edges. This **can't** be removed completely but *achromatization* can be achieved by selecting the right type of glasses.

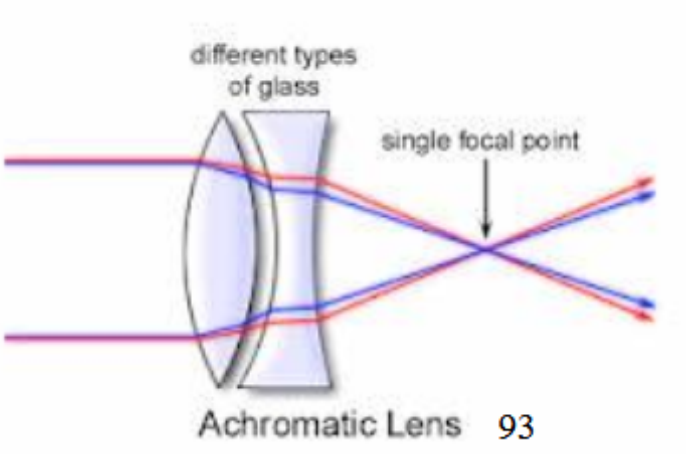



**Figure 18:** Achromatization

#### 2.4.3. Photometric camera model

We want to convert the object radiance into a pixel grey level. This is a 2-step process with our idea of the image plane. We first project the radiance of the object to the image, then this image irradiance projects to a pixel in a grey level. The **cos4** law governs the radiance. The more an object is of-axis from the center by an angle $\theta$ the more it reduces its irradiance in the pixel. This materializes with **natural vignetting**[3] (like the old instagram filter).

$$I = R\frac{A_l}{f^2}\cos^4\theta$$

This will get converted in a grey pixel with:

$$pix = gI^\gamma + d$$

With the $g$ of the camera, $d$ the dark reference of the camera. The gamma $\gamma$ is a non-linear relationship which changes the mid-tone without really changing the pure black or white. A value lower than 1 makes the image lighter and vice-versa. Nowadays, cameras have become quite linear meaning that $\gamma = 1$.

## 3. Sampling and Quantisation

We live in an analog world but our computers and digital world is finite. So we have to sample (Discretize points) and quantize (attribute them a finite value).

From what we have seen and learned before, we can trick the human eye to see a higher quality image than it is actually. For example the display on a phone has a RGB pixel ratio as 2:1:1 because the human is less sensible to green. We call

[3] optical vignetting appears due to an obstruction of the lens by an element

this grid the **bayer filter**. Other representation exist which can be more exotic, e.g.: hexagonal (more isotropic, similar structure as in the retina, no connectivity ambiguities).

We have quantization where we usually talk about levels. Typically a n bits representation has $2^n$ levels. We can also apply fancier quantization scheme instead of simple linear one. The most classic representation is the RGB888 (1 byte per pixel for each color).

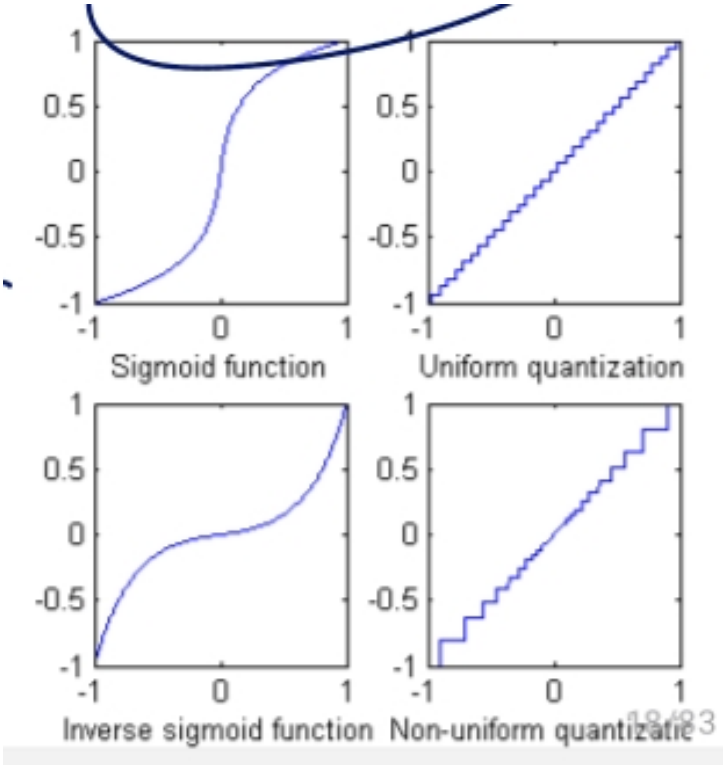
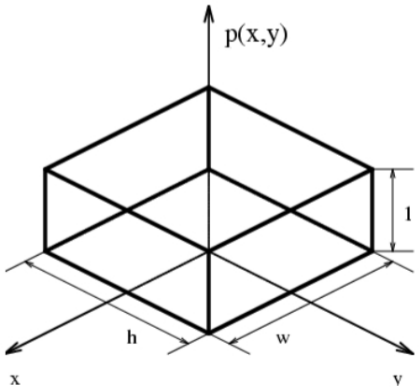



**Figure 19:** Various quantization scheme

**3.0.0.1. HDR** High Dynamic Range (HDR) is a technology where we have various luminance on the screen giving really good contrast in dark and bright area. Usually, we combine multiple photos with various gains and then recombine it all together with the various exposure.

## 3.1. Sampling

### 3.1.1. Integrate brightness over the cell window




$$o(x', y') = \iint i(x, y)p(x - x', y - y')dxdy$$

This is *convolution:* $i(x, y) * p(-x,-y)$

**Figure 20:** Integrating over a pixel cell

This forms a convolution. Convolution are **linear**, $f_1 \to g_1 \quad f_2 \to g_2 \Rightarrow af_1 + bf_2 \to ag_1 + bg_2$ , and **shift-invariant**, $f(x, y) \to g(x, y) \Rightarrow f(x - a, y - b) \to g(x - a, y - b)$.

Convolutions are also commutative and associative as proven in the Fourier domain.

$$\mathfrak{F}(u) = \int_{-\infty}^{\infty} f(t)e^{-2i\pi xu}dx$$

For the 2D case, we have $e^{-2i\pi(ux+vy)}$ which has for Euler form $\cos(2\pi(ux + vy)) + i\sin(2\pi(ux + vy))$. Now, the $u$ and $v$ represents the frequency along the x and y dimensions. To obtain the $\lambda$ of the signal we need to take the norm of the frequency for both axis inverted:

$$\lambda = \frac{1}{\sqrt{u^2 + v^2}}$$

As in 1D, we can do Fourier transform to decomposition the image in a bunch of frequencies. As in 1D, we have a complex results where the magnitude is $|\mathfrak{F}(u, v)| = \sqrt{\mathbb{R}(\mathfrak{F}(u, v))^2 + \mathbb{I}(\mathfrak{F}(u, v))^2}$ and the phase angle $\angle\mathfrak{F}(u, v) = \arctan(\mathbb{I}(\mathfrak{F}(u, v))/\mathbb{R}(\mathfrak{F}(u, v)))$.

$$\mathfrak{F}(u, v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)e^{-2i\pi(xu+yv)}dxdy$$

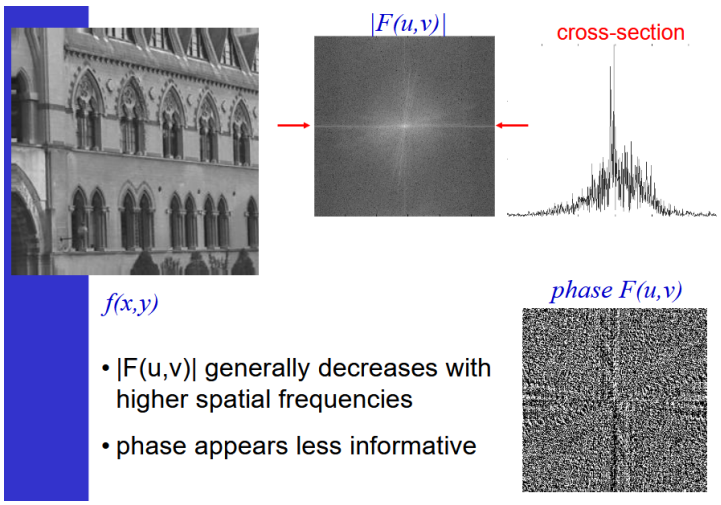



**Figure 21:** Fourier decomposition of images

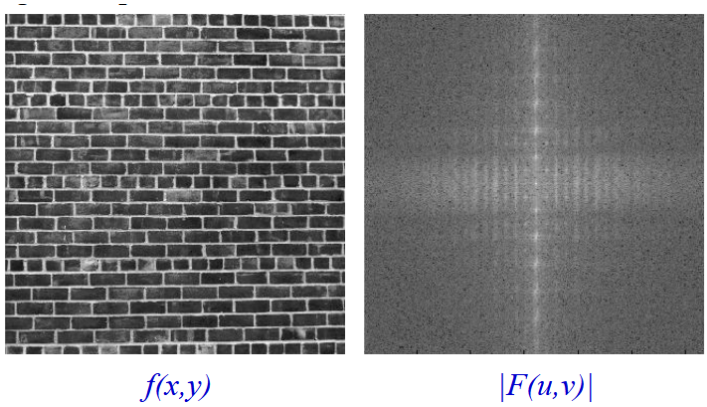



**Figure 22:** Repetitive patterns in an image creates repetitions in frequency domain

Removing patterns peak in frequency allows to remove periodic background.

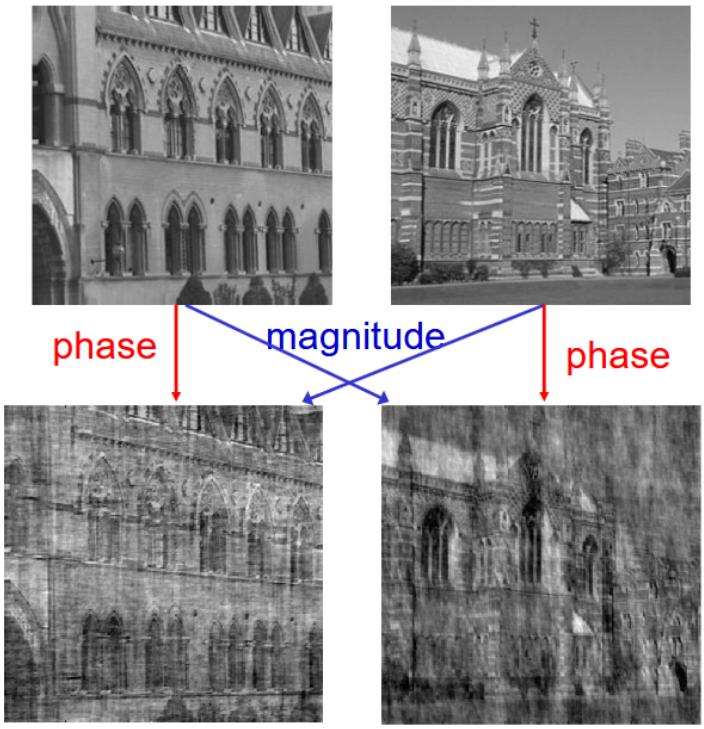



**Figure 23:** Mixmatch of phase and magnitude

#### 3.1.1.1. Properties of spatial-frequency domain

| Spatial domain | Frequency domain |
|---|---|
| Real | Real part → even; Imaginary part → odd |
| Real,even | Real, even (cosine) |
| Real,odd | Imaginary, odd (sine) |

Important to remember that a convolution in spatial domain is a multiplication in the frequency domain and vice versa.

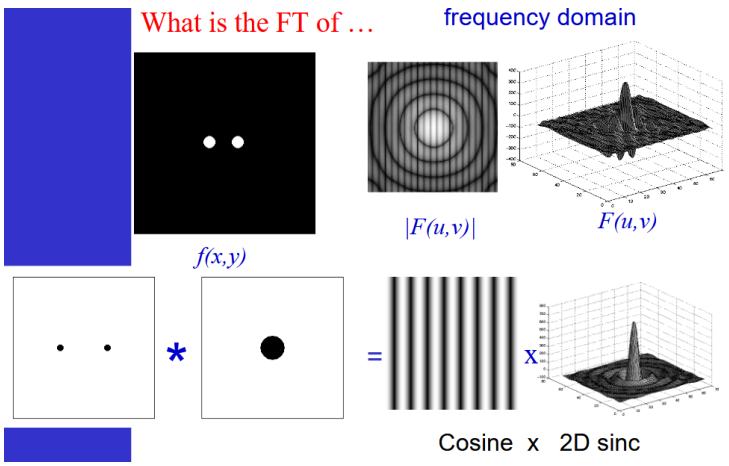



**Figure 24:** FFT approach

We have the base image (the dot) and the the transfer function, MTF R, which contains the amplification and phase shift information to every frequency in the input image I.

#### 3.1.1.2. Integrating with the convolution

Re-using the integration over a cell window:

$$o(x', y') = \int\int i(x, y)p(x - x', y - y')dxdy \leftrightarrow i(x, y) * p(-x, -y)$$

We can then transform this $p(x, y)$ round-point (as seen in the previous figure) in a $P(u, v)$

$$P(u, v) = \int_{-\infty}^{\infty} e^{-i2\pi(ux+vy)}p(x, y)dxdy \tag{2}$$

$$= wh\left(\frac{\sin(\pi wu)}{\pi wu}\right)\underbrace{\left(\frac{\sin(\pi hv)}{\pi hv}\right)}_{\text{sinc}} \tag{3}$$

The sinc function is predominantly low-pass **and** non-causal (no phase shift) and it has phase reversal.

## 3.2. Aliasing

This corresponds to a 2D Dirac train convolution on the grid like:

$$f(a, b) = \int\int f(x, y)\delta(x - a, y - b)dxdy$$

The multiplication with the 2D pulse train with a spacing of distance w and h:

$$\sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty}\delta(x - kw, y - lh) \tag{4}$$

$$\longleftrightarrow \frac{1}{wh}\sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty}\delta(x - k\frac{1}{w}, y - l\frac{1}{h}) \tag{5}$$

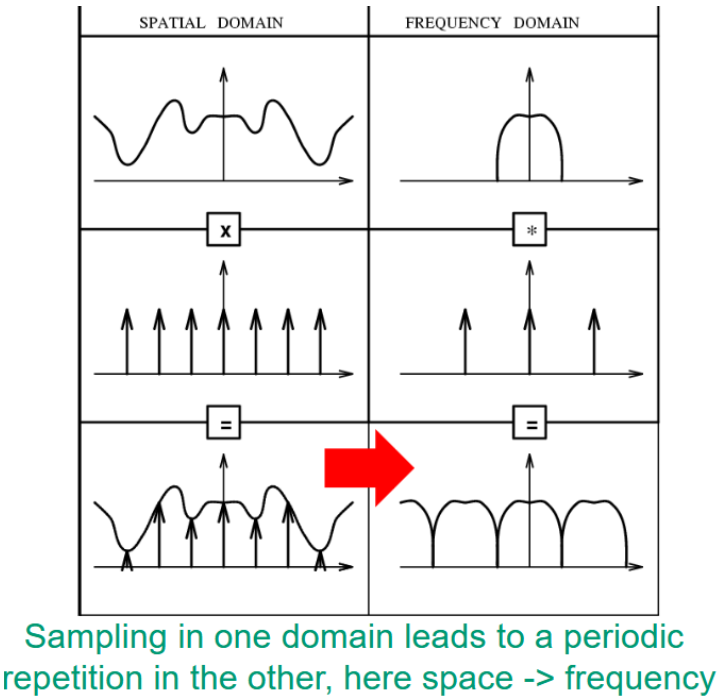



**Figure 25:** Basic sampling theorem

Also, we need to do a Discrete Fourier Transform, DFT, to discretize the space and frequency. We can also use FFT to bring the complexity of the algorithm from $N^2$ to $N \log_2 N$.

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} F(k,l) e^{2\pi i \left(\frac{km}{M} + \frac{ln}{N}\right)} \tag{6}$$

$$\longleftrightarrow F(k,l) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n) e^{-2\pi i \left(\frac{km}{M} + \frac{ln}{N}\right)} \tag{7}$$

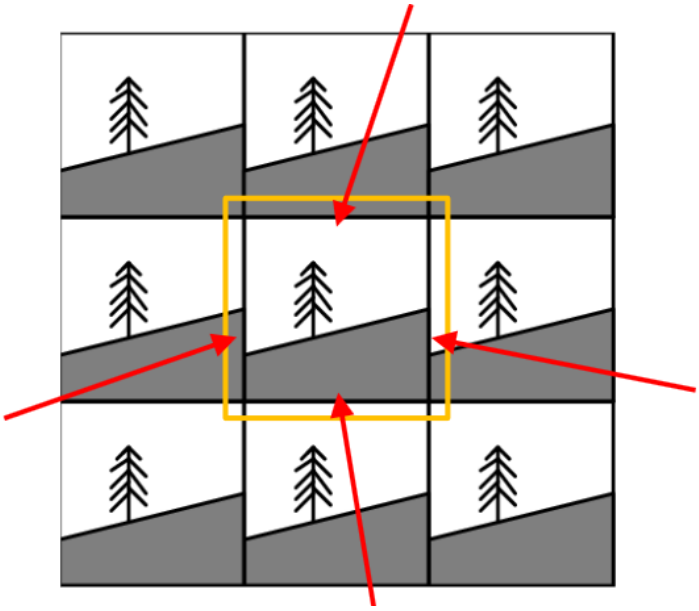


**Figure 26:** Periodicity assumed in both domains, might introduce false high frequencies at image boundaries

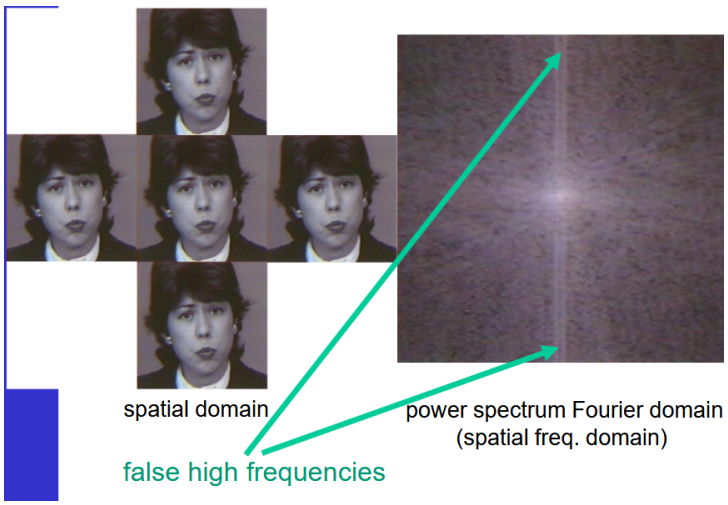
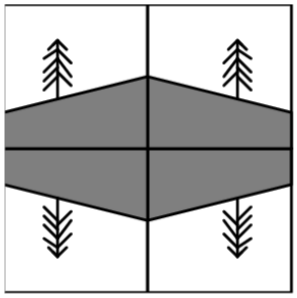



**Figure 27:** We can have false high frequencies due to sampling



*Instead of taking the DFT of the original image, one can take it of the repeated one, as shown on the left.*
*Periodically extending it avoids the creation of high frequencies near the boundaries.*
*This is the discrete cosine transform or DCT (only cosines as the input is real and even)*

**Figure 28:** Solution on periodicity

Again, the Shannon theorem is important in 2D as well such that the sampling distance $w, h$ along the $x, y$ axis must follow:

$$w \leqslant \frac{1}{2u_b} \qquad h \leqslant \frac{1}{2v_b}$$

## 4. Enhancement and feature detection

Strong from the knowledge of the previous section, we can do some spectral filtering. For example, if we have some white noise in an image we could apply a low pass filter on it to already remove a big part of the noise while losing a marginal part of the actual picture information. Usually, this will sorts of blur the image because sharp edges are like high-frequency information.

### 4.1. Enhancement - better look of image

Typically, HDR is a form of enhancement for better contrast. A cheaper way of doing HDR is doing histogram equalisation which allows for better usage of the full dynamic range available (a bit like gamma correction I think).

### 4.1.1. Histogram Equalisation

This will makes the image more vibrant if we exploit more of the dynamic range, we **redistribute** the intensities through a **mapping that keeps their relative order**. Ideally, we make the histogram **flat**.

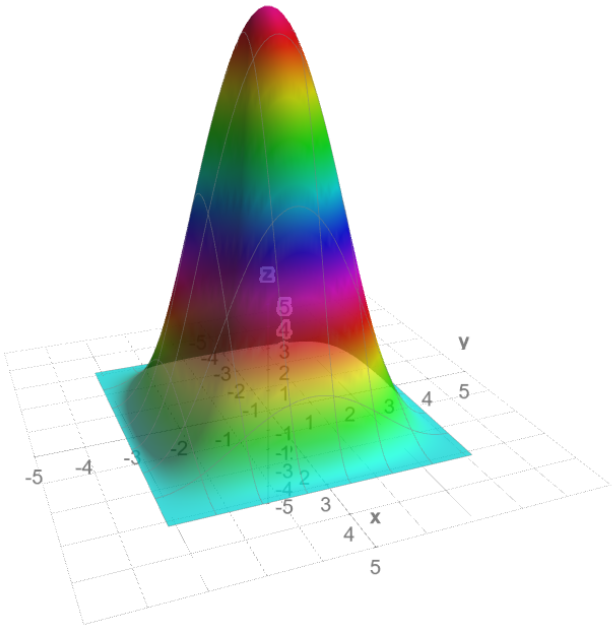



**Figure 29:** We tune the Cumulative Intensity Probability - or CPD to a flat line

We will compress the flat parts of the CPD, and expand the steepest one. But the main issue, since we move bins, we want have a flat histogram, we will just spread the bins more equally due to the **discrete nature** of the input. On top, neigboring pixels won't always have the same difference of intensities after the remapping (less problematic though).

### 4.1.2. Deblurring or Unsharp masking

If we deblur we actually increase the image sharpness. It's *simple, linear, image independent and effective*.

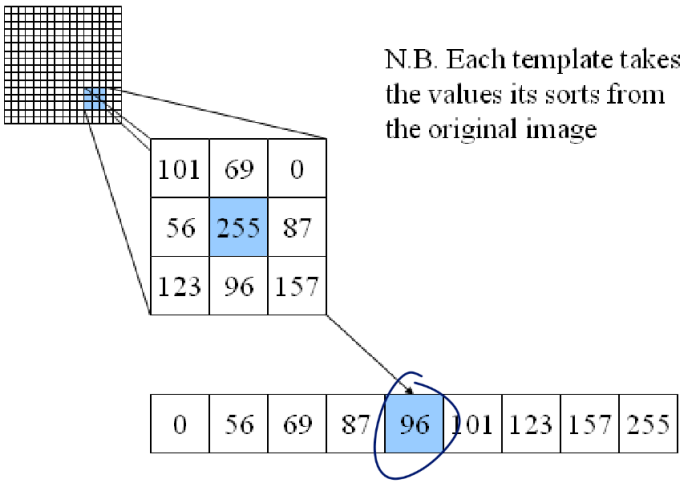



**Figure 30:** Imagine a vertical cut of image where x-axis is the pixel number and y-axis the grey-level

We see that we, indeed, make the edge steeper. Moreover, the difference between the original-smooth makes a good fit for second order derivative. This is a bit like running a diffusion process backwards.

The little over and under-shoots at the edge even magnifies the contrast (Mach band effect).

### 4.1.3. Inverse Filtering

Now, we shift gears and go into the frequency. We have the frequency function that blurs the image with $B(u, v)$. To undo it we apply $B'(u, v)$ such that $B'(u, v)B(u, v) = 1$. Two danger here:

1. What happens for $B(u, v) = 0$
   - Use a constant at the denominator –> Spurious high-frequencies :'(
   - $B'(u, v) = B(u, v)/(B(u, v)^2 + C)$ –> Much better ! Wiener-like reduces C for larger SNR and vice versa.
2. For low $B(u, v)$ after noise was applied –> may over-fit the noise…

### 4.1.4. Noise Suppression

**4.1.4.1. Low pass** As introduced, we can use a low-pass filter for noise suppression. The main problem with applying this rectangular (actually a circle) window is the rippling due to the sinc function as seen here

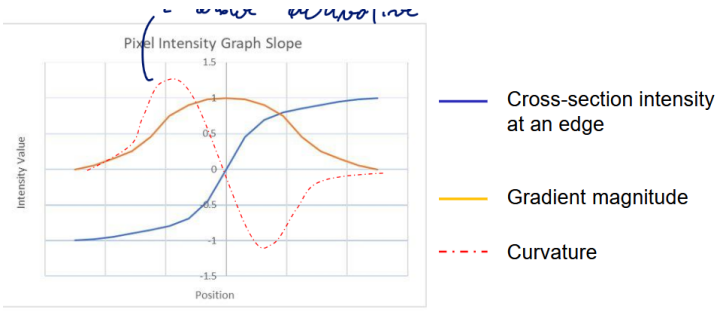



**Figure 31:** Rippling effect

## 4.2. Features - edge and corner detections

**ABOVE SHOULD BE THE START OF CHAPTER ABOUT ENHANCEMENT AND EDGES**

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**4.2.0.1. Convolution filters** It is based on the insight that noise affects high frequencies the most due to their low SNR. To build a low-pass filter, we will use an averaging kernel such:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} * \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{\text{separable filter}}$$

The two $(1, 1, 1)$ vectors can be seen in the *frequency* domain as $1 + 2\cos(2\pi u)$ for the horizontal domain or $1 + 2\cos(2\pi v)$ in the vertical one. The intuition for this is to look at the vector as a 1D image centered around the middle value. Finally, the convolution can be simplified in the frequency domain as:

$$(1 + 2\cos(2\pi u))(1 + 2\cos(2\pi v))$$

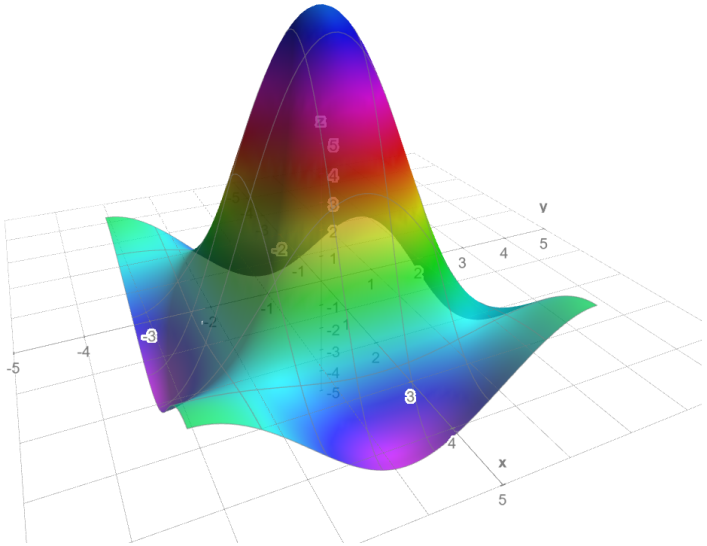



**Figure 32:** Representation in 3D of the function

The function is purely real and low-pass with some phase reversal! We can again derivate something similar for a 5x5 kernel:

$$(1 + 2\cos(2\pi u) + 2\cos(4\pi u))(1 + 2\cos(2\pi v) + 2\cos(4\pi v))$$

With this one, the phase reversal is even more significant, the higher the frequency the more it looks like a sinc function. This creates more ripples and disturbing effects. Hard to get no ripple and a real low-pass for higher order functions.

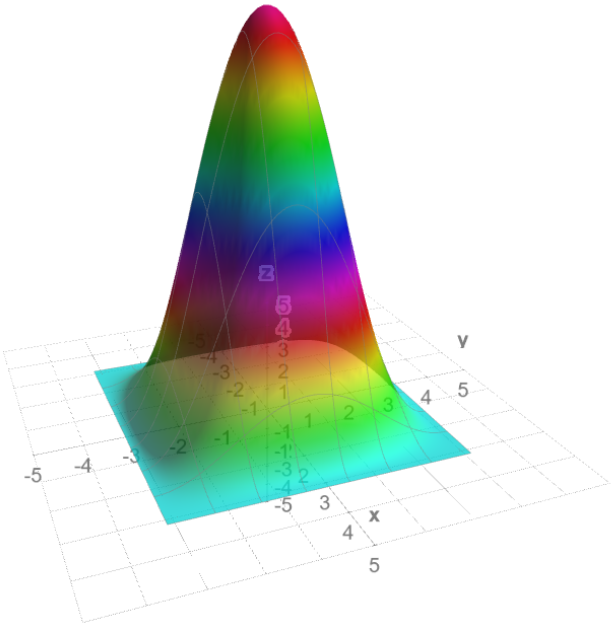
From all those experiments we can draw 3 conclusions:

1. Separable filters can be implement more efficiently
2. Better to go in the frequency domain for large kernel
3. Keep it simple, mask of boolean or power of 2 are cheap!

### 4.2.1. Binomial filters

They are here to solve the ripple and low-pass issue of the convolution filters presented before. This guarantees no phase reversal.

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} * \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \longleftrightarrow (2 + 2\cos(2\pi u))(2 + 2\cos(2\pi v))$$




{width=50%}

Note that noise removal linear filters will exhibit blurring due the fact they cannot difference noise and actual information.

### 4.2.2. Median

Instead of averaging over using a kernel of ones, we take the median - the middle point of neighborhood of pixel. We don’t create a new grey level, we take the one in the middle.

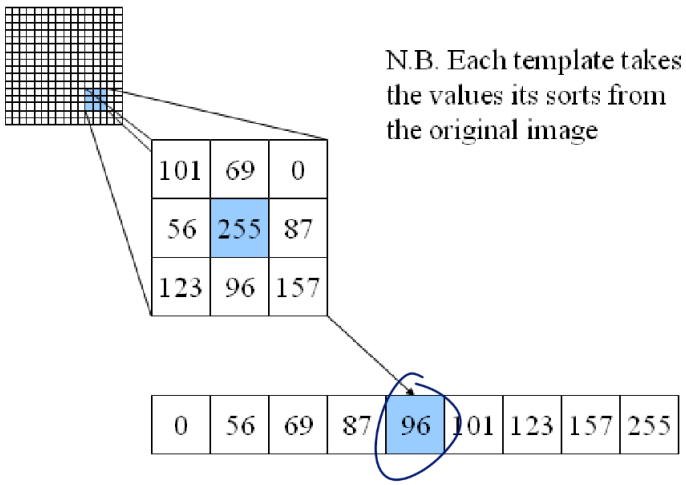



**Figure 33:** Median filters principle - credits: southampton.ac.uk

One advantage of a median filter is the fact it is robust against outliers and it will preserve the contrast instead of smooching out all the values and making the image more blurry.

- Discard outliers
- Preserve discontinuities

But, it will loose in details (especially small areas). Moreover, give a patchy effect instead of a smooth uniform color.

#### 4.2.3. Anisotropic filters - all-in-on of filtering

This filter does:

1. Binomial smoothing between edges
2. Unsharp masking at edges

For this, we need to cover edge detection

**Table 5:** Summary table of the enhancement procedures

| Operation name | Description | Linear ? |
|---|---|---|
| Histogram equalisation | Re-sort histogram to create a more linear Cumulative Probability Density | V |
| Unsharp mask | | V |
| Inverse Filtering | Realize the opposite operation. Need to carefully handle low SNR with a constant. | V |
| Low pass | Removes noise since it affects more high-frequencies due to low SNR | V |
| Binomial | Like low-pass but avoids phase reversal | V |
| Median | Take the middle point instead of the mean | X |
| Anisotropic | Binomial between edges - Unsharp at edge | X |

### 4.3. Enhancement and Feature Detection

#### 4.3.1. Edges

> **Definition:**
>
> Edge correspond to image locations with strong intensity changes due to:
>
> 1. Reflectance
> 2. Orientation of shape normals (its surface) - the object's edge
> 3. Illuminations (shadow)
>
> The hardest challenge in edge detection is linking up all the pixels into a good line drawing.

The methods cover here will be based on the gradient which gives the direction of the steepest change along as its magnitude being the rate of change:

$$\text{gradient } = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) \qquad \text{magnitude } = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

2 methods will be covered using **isotropic** operators[4]:

1. Locating high intensity gradient magnitude

[4] there exists other (and better) operators but too complex for this class

2. Locating inflection points in the intensity profiles

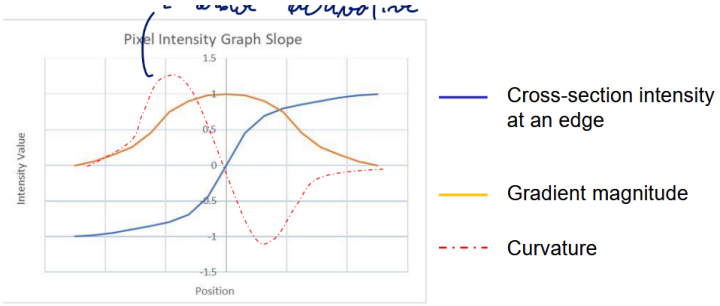



**Figure 34:** With curvature being a sort of second order derivative

### 4.3.2. Gradient

The idea is to:

1. measure the gradient **magnitude** of all the steep slopes in the image. This operator is isotropic since there are equal contribution of the x and y direction.
2. Measure the angle $\theta$ that maximizes $\frac{\partial f}{\partial x'} = \frac{\partial f}{\partial x}\cos(\theta) + \frac{\partial f}{\partial y}\sin(\theta)$ this yields $\theta_{xtr} = \tan^{-1}\left(\frac{\partial f}{\partial y}/\frac{\partial f}{\partial x}\right)$

The gradient is a non-linear operator of $\partial/\partial x$ which are linear, shift-invariant ! So this can be implemented using a convolution, the **Sobel masks** which are separable and cheap to implement. The Sobel masks are a bit like a binomial filter but with a derivative in the middle. The idea is to have a strong response for high frequencies (edges) and low response for low frequencies (flat areas).:

$$\frac{\partial}{\partial x} = (-1,0,1) * (1,2,1)^\top = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix} \qquad \frac{\partial}{\partial y} = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

From their outputs, we take the square root of the sum of their squares, take the arctan which will give us the orientation of the edge!

Again we can move to the frequency domain where we remember we can see the vector as a 1D image centered around the middle value. The Fourier transform of the Sobel mask is:

$$(-1,0,1) \longleftrightarrow 2i\sin(2\pi u) \qquad (1,2,1) \longleftrightarrow 2 + 2\cos(2\pi u)$$

We see that it is pass band along the u-dir (red on the graph) and low-pass along the v-dir (green on the graph).

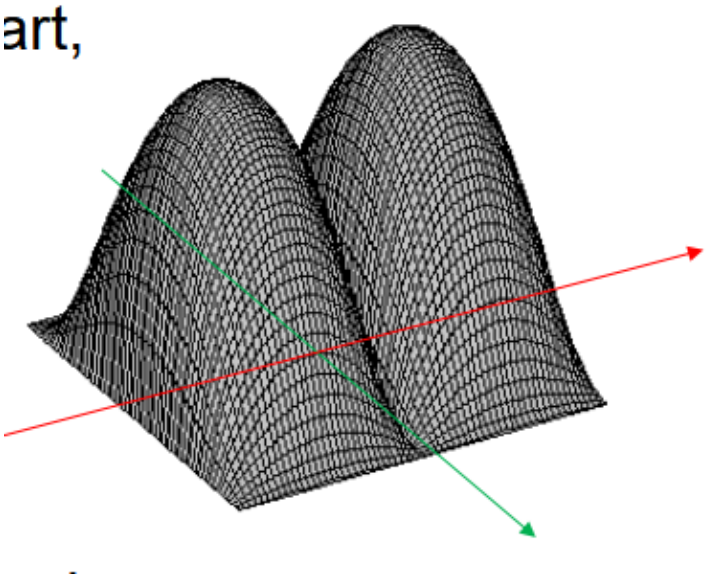



**Figure 35:** Function in 3D

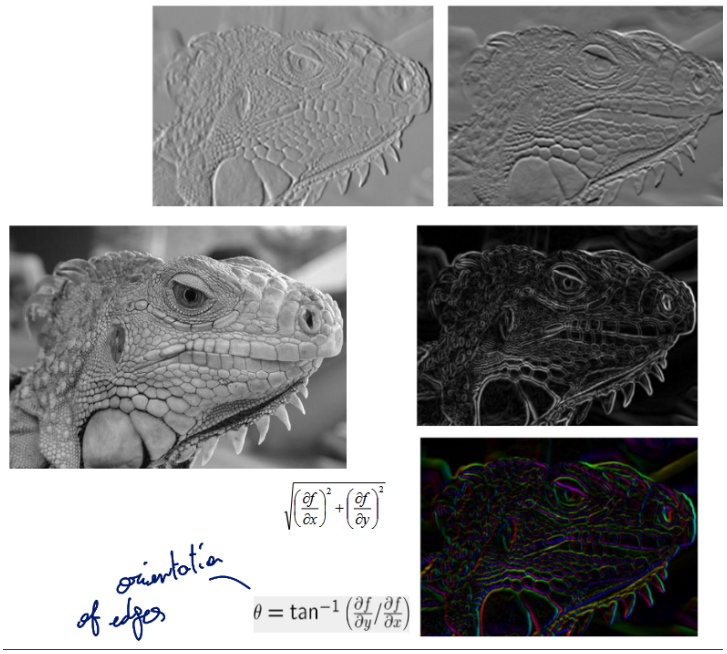



**Figure 36:** Combining the two (top left: x-dir, top right: y-dir)

It is a great technique to map individual pixel edges, but cannot be perfect to edge detection. There can be gap, several pixels thick, weak or salient edges, and so on. But Sobel masks are the optimal 3 x 3 convolution filters with integer coefficients for step edge detection.

### 4.3.3. Zero-crossing

First, we want to do edge thinning, to reduce a several pixel thick edge to only one continguous line. Once we found the gradient, we want to look at the direction giving the maximum gradient and interpolate to get these values. We can also apply the hysteresis thresholding to link up the edges. The idea is to have a high threshold for strong edges and a low threshold for weak edges. We then link the weak edges to the strong ones if they are connected.

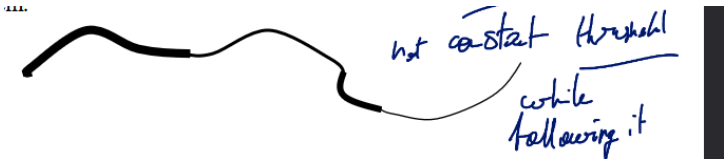



**Figure 37:** Canny edges

To do zero-crossing detection, we need to use second order derivative and make it equal to 0 to find inflexion points. This is linear + shift invariant (convolution) and isotropic (equal contribution of x and y). The Laplacian is sensitive to noise, so we often add a smoothing step (Gaussian) before applying the Laplacian. This is called the Laplacian of Gaussian (LoG) operator.

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

We can also use the Difference of Gaussian (DoG) which is a good approximation of the LoG and is much faster to compute. The idea is to take the difference between two Gaussian blurred images with different standard deviations.

$$G_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad G_2 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -12 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

This DoG method will give us a closed-contour as it is based on the second order derivative. The zero-crossing will give us the edge location. The main issue is that it is not very good at linking up edges and can be quite sensitive to noise.

#### 4.3.4. Harris corner detection

This method will distinguish between homogeneous areas, edges and corners. The idea is to look at the eigenvalues of the second moment matrix of the gradient. If both eigenvalues are small, we have a homogeneous area. If one eigenvalue is large and the other is small, we have an edge. If both eigenvalues are large, we have a corner. The second order moment matrix is given by:

$$\begin{pmatrix} \left(\frac{\partial f}{\partial x}\right)^2 & \frac{\partial f}{\partial x}\frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial x}\frac{\partial f}{\partial y} & \left(\frac{\partial f}{\partial y}\right)^2 \end{pmatrix}$$

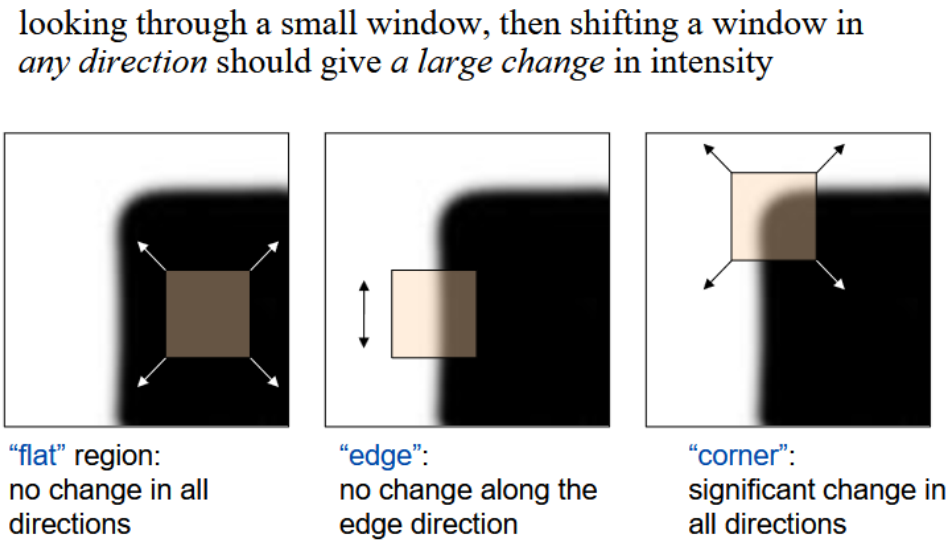



**Figure 38:** Principle of Harris

The idea is to find the directions of minimal and maximal change in the image. We move the kernel by a small amount (u,v) where our $w$ window function can be either a box or a Gaussian. We then look at the change in intensity which is given by:

$$E(u,v) = \sum_{x,y} w(x,y)[f(x+u, y+v) - f(x,y)]^2$$

$$E(u,v) \approx [u,v]M \begin{bmatrix} u \\ v \end{bmatrix} \qquad \text{with } M = \sum_{x,y} w(x,y) \begin{pmatrix} \left(\frac{\partial f}{\partial x}\right)^2 & \frac{\partial f}{\partial x}\frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial x}\frac{\partial f}{\partial y} & \left(\frac{\partial f}{\partial y}\right)^2 \end{pmatrix}$$

To analyze the harris detector, we use the iso-lines $= det - ktrace^2 = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$. Where $\lambda_1$ and $\lambda_2$ are the eigenvalues of the matrix $M$ and represent a change in one or the other direction. If the iso-line is positive, we have a corner. If it is negative, we have an edge. If it is zero, we have a homogeneous area. The main issue with this method is that it is not very good at distinguishing between edges and corners in the presence of noise.

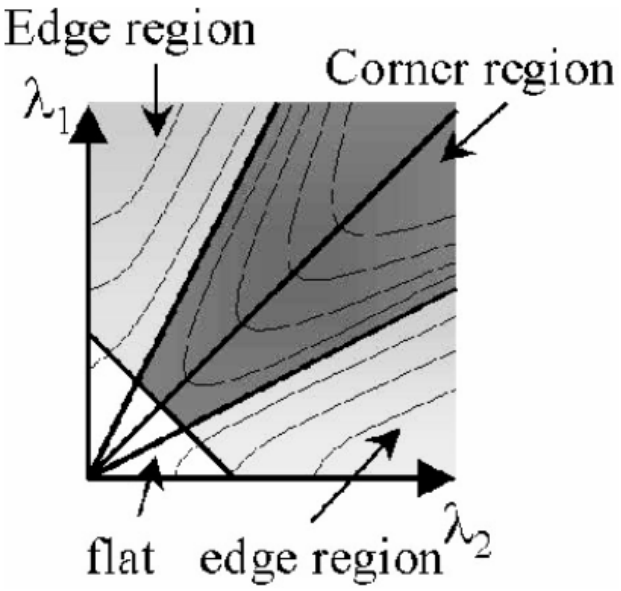



**Figure 39:** Iso-lines for Harris

## 5. Image decomposition

Images can be projected in various orthogonal basis, (pixels, fourier domain, ...). We can also do a rotation for example, but the most important thing is that we use **unitary transformations** $\rightarrow U^*U = I$.

### 5.1. PCA

We want to reduce the dimensionality of the data while preserving as much variance as possible. We can do this by projecting the data onto the eigenvectors of the covariance matrix. The eigenvectors with the largest eigenvalues will be the principal components. This is a linear transformation that preserves the distances between points in the original space. It is an **unitary transformation** but **data dependent**. We want to reduce the dimension of the data while keeping as much information as possible.

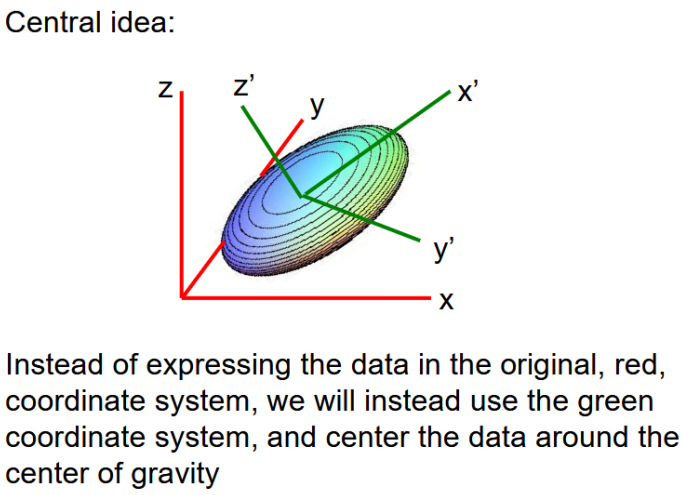



**Figure 40:** Central idea of PCA

The Shannon entropy $H(x)$ quantifies the amount of entropy in a data set with a distribution P. This fact is interesting, because most pictures only span a sub-set of the total possible images. For example, natural images have a lot of structure and regularity, which means that they have a low entropy compared to random noise. This is why we can use PCA to reduce the dimensionality of the data while preserving most of the information.

$$H(x) = -\sum_i P(x_i) \log P(x_i)$$

This issue with the lack of spanning in high-dimensional space is often referred to as the **curse of dimensionality**. PCA is a solution to this problem. Some intuitive explanation: 1D: $(r+\delta)/r \approx 1 + \delta/r$; 2D: $(r+\delta)^2/r^2 \approx 1 + 2\delta/r$; D: $(r+\delta)^D/r^D \approx 1 + D\delta/r$. So the volume of the space grows exponentially with the dimension, which means that we need an exponentially large number of samples to cover the space. PCA helps us to reduce the dimensionality of the data while preserving most of the information.

For the PCA, we use the Pearson correlation coefficient. We need to be careful with it as it only indicates linear correlation between two variables!

$$p_{ij} = \frac{cov(x,y)}{\sigma_i \sigma_j} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2}\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}$$

We first find the highest component with the maximum of correlation, then we find the second one with the lowest correlation. We can then apply a rotation in a high dimensional space.

By working around the mean, we benefit from parserval-equality which indicates the variances in $x_i$ will be the same in $z_j$. This indicates a redistribution of the energy.

$$\sum_{i=1}^p \sigma_i^2 = \sum_{j=1}^p \tilde{\sigma}_j^2$$

### 5.1.1. Method

With $x$ a vector of $p$ variables. We first look for a linear combination $c_1^\top x$ which has the maximum variance. We then look for a second linear combination $c_2^\top x$ which has the maximum variance and is uncorrelated with the first one. We repeat this process until we have $p$ linear combinations. The first few linear combinations will capture most of the variance in the data, which is why we can reduce the dimensionality of the data by keeping only the first few principal components.

By relying on the Gaussian distribution postulat, we can derive a covariance matrix. Note, we need to center the data around the mean first. We need to maximize so $var(c_1^\top x) = \sum c_1^\top x (c_1^\top x)^\top = c_1^\top \sum (xx^T) c_1 = c_1^\top C c_1$ with $C$ the covariance matrix. And we make sure that $c_1^\top c_1 = 1$ to avoid trivial solution.

$$C = \begin{bmatrix} \mathrm{E}[(X_1-\mu_1)(X_1-\mu_1)] & \mathrm{E}[(X_1-\mu_1)(X_2-\mu_2)] & \cdots & \mathrm{E}[(X_1-\mu_1)(X_n-\mu_n)] \\ \mathrm{E}[(X_2-\mu_2)(X_1-\mu_1)] & \mathrm{E}[(X_2-\mu_2)(X_2-\mu_2)] & \cdots & \mathrm{E}[(X_2-\mu_2)(X_n-\mu_n)] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{E}[(X_n-\mu_n)(X_1-\mu_1)] & \mathrm{E}[(X_n-\mu_n)(X_2-\mu_2)] & \cdots & \mathrm{E}[(X_n-\mu_n)(X_n-\mu_n)] \end{bmatrix}$$

Sample = $(X_1, X_2, X_3, \ldots, X_n)$

**Figure 41:** Covariance matrix

We can then use the Lagrange multiplier with the constrained mentioned ealier:

$$\mathcal{L}(c_1, \lambda) = c_1^\top C c_1 - \lambda(c_1^\top c_1 - 1)$$

$$\frac{\partial \mathcal{L}(c_1,\lambda)}{\partial c_1} = 2Cc_1 - 2\lambda c_1 = 2c_1 \underbrace{(C - \lambda I)}_{\text{eingevalue problem}} = 0 \qquad \frac{\partial \mathcal{L}(c_1,\lambda)}{\partial \lambda} = c_1^\top c_1 - 1 = 0$$

So $c_1^\top C c_i = c_1^\top \lambda c_1 = \lambda c_1^\top c_1 = \lambda$ with $\lambda$ as large as possible. Decorrelation and orthogonality are the same thing in this case since we are in a Euclidean space. So $c_2^\top C c_1 = c_2^\top \lambda c_1 = \lambda c_2^\top c_1 = 0$.

Proof for $k = 2$

Maximize $c_2^T C c_2$ being uncorrelated with $c_1^T x$

$$\mathrm{cov}\left[c_1^T x, c_2^T x\right] =$$

$$c_1^T C c_2 = c_2^T C c_1 = c_2^T \lambda_1 c_1 = \lambda_1 c_2^T c_1 = \lambda_1 c_1^T c_2$$

$$= 0$$

Thus uncorrelatedness becomes

$$c_1^T C\, c_2 = 0,\ c_2^T C\, c_1 = 0,\ c_1^T c_2 = 0,\ c_2^T c_1 = 0$$

i.e. $c_2$ is orthogonal to $c_1$

**Figure 42:** Proof of orthogonality between $c_1$ and $c_2$

We can again use a Lagrange multiplier to ensure decorrelation:

PCA: algorithm

Then, using λ, ϕ as Lagrange multipliers

$$c_2^T C c_2 - \lambda\left(c_2^T c_2 - 1\right) - \phi\, c_2^T c_1$$

Differentiation w.r.t. $c_2$ gives

$$C c_2 - \lambda c_2 - \phi c_1 = 0$$

Multiplication on the left by $c_1^T$ gives

$$c_1^T C c_2 - \lambda c_1^T c_2 - \phi\, c_1^T c_1 = 0$$

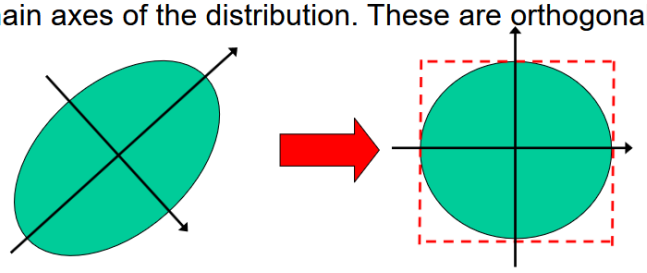
Thus $\phi = 0$

Therefore, $C c_2 - \lambda\, c_2 = 0$ , i.e. $\left(C - \lambda I_p\right) c_2 = 0$

Again, $\lambda = c_2^T C c_2$ , so select 2nd largest $\lambda$

**Figure 43:** Decorrelation

The decorrelated projections correspond to the main axes of the distribution. These are orthogonal.



**Important note:** principal components are guaranteed to be independent only if the data set is jointly normally distributed (i.e. ellipsoid-shaped).

Through the covariance matrix we only extract moments up to 2nd degree, and needed to specify such Gaussian distribution

**Figure 44:** Important remark

### 5.1.2. Usage

- Classification of crops: using Near-infrared, Red, Green data. We see some correlation between red and green + near-infrared. We can compress the data to roughly 60% while retaining similar performance.
- Culture and heritage: reveal details that wasn't visible before. Quickly find the most important features in the data.
- Image compression: same density in neihboring pixels. Issue is that image of N pixels =$N^2$ base. So $N^2 * N^2$ for the covariance matrix. PCA allows us to reduce the dimensionality of the data while preserving most of the information. We can then use a lossy compression algorithm to further reduce the size of the data.
  - We can use the **eingenfaces** of the covariance matrix as a basis for the image. We can then project the image onto this basis and keep only the first few components to reduce the size of the data. This is called **eigenface**

method. We form a human face base using a base of already existing faces. (do not forget to substract the mean !)

#### 5.1.3. Limitations

As described before, the Pearson correlation coefficient is only for linear correlation. Moreover, the PCA is based on the assumption of a Gaussian distribution.

A solution is the Interesting Component analysis where we find multiple “interesting” components. Or we can use the Linear Discriminant Analysis (LDA) which is a supervised method that finds the linear combination of features that best separates two or more classes of objects or events. It is based on the idea of maximizing the between-class variance while minimizing the within-class variance. This approach is better for non Gaussian distribution.

There is also the support vector machine which look at an hyper plane to find the best separation between classes.

PCA looks like the Harris corner detector but a few things chenge:

- No substraction by mean in Harris
- No normalization in Harris

## 6. Surface features color and texture

### 6.1. Color

It is important to distinguish between two important spectre of the color:

1. Radiometry: physical measurement of the light (radiance, irradiance, reflectance, ...)
2. Photometry: how the human eye perceives the light (brightness, colour, ...)

A light $l$ with a spectral composition $c(\lambda)$ and the human that perceives it with a sensitivity $h(\lambda)$, with the constant $k = 683$ lm/W. Ofc, we can span only the human visible spectrum which is roughly between 400 and 700 nm. The perceived brightness is then given by the following equation:

$$l = k \int_{\lambda=0}^{\infty} c(\lambda)h(\lambda)d\lambda$$

Human perception of color can be summarized with Brightness, Huhe, and saturation.

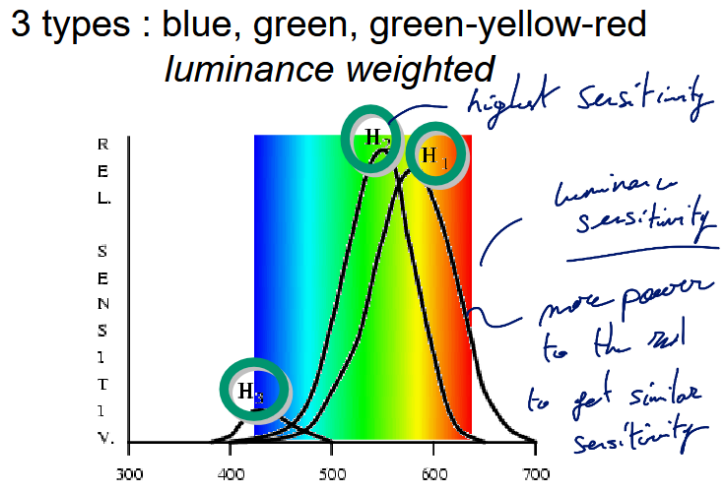



**Figure 45:** Retinal cones of a human; occupancy S:5-10, M:30, L:60

So the human has 3 sensitivity curves and not just one from the equation aforementioned. We have 3 responses for the full spectrum. So if 2 sources are observed with the same response for the 3 cones, they will be perceived as the same color. This is called **metamerism**.

### 6.1.1. Commission Internationale de l'Éclairage (CIE)

They standartized the color space with 3 primaries:

1. Red: 700 nm
2. Green: 546.1 nm
3. Blue: 435.8 nm

With this, we can match $m_j$ a source $C(\lambda)$ with a combination of the 3 primaries $P_j(\lambda)$ such that:

$$C(\lambda) = \sum_{j=1}^{3} m_j P_j(\lambda)$$

We can derive the following:

$$R_i(C) = \int \sum_{j=1}^{3} m_j P_j(\lambda) H_i(\lambda) d\lambda = \sum_{j=1}^{3} m_j \underbrace{\int P_j(\lambda) H_i(\lambda)}_{l_{i,j} \text{ Sensitivity of a primary can be determined offline}} d\lambda$$

This makes a really simple system of 3 linear equations for the human eye. If we use different primaries $L'$, we can find a new set of coefficients $m'$ such that:

$$\sum_{j=1}^{3} m_j l_{i,j} = Ri \qquad LM = R \qquad L'M' = R \quad m' = L'^{-1} Lm$$

**6.1.1.1. Tristimulus value** As white is the combination of all the colors, we can use the white point as a reference for the color. We can then define the tristimulus value as:

$$T_j = \frac{m_j}{w_j} \qquad T_1 = T_1 = T_3 = 1 \quad \text{for white}$$

$$R_i(C_\lambda) = H_i(\lambda) \sum_{j=1}^{3} m_j \, l_{i,j} = \sum_{j=1}^{3} w_j \, l_{i,j} \, T_j(\lambda)$$

for the CIE primaries.

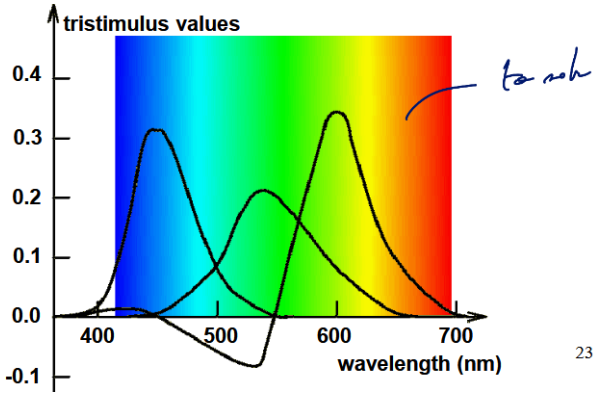



**Figure 46:** Spectral matching curves

We notice negative colors which cannot be produced by a combination of the primaries. This is a problem for the display of colors. Moreover, for some colors, a triple of primaries cannot represent it! However the tristimulus values still contain brightness information, we can normalize to only get chromatic information:

$$t_j = \frac{T_j}{\sum_{j=1}^{3} T_j} \qquad t_1 + t_2 + t_3 = 1$$

So now, 2 chromaticity coordinates are enough to represent the color (saturation and hue). This forms the XYZ coordinates of the CIE, with X being a mix of red/green cones, the Y the luminance and Z blue light primarly. The mapping matrix is

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0.490 & 0.310 & 0.200 \\ 0.177 & 0.813 & 0.011 \\ 0.000 & 0.010 & 0.990 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

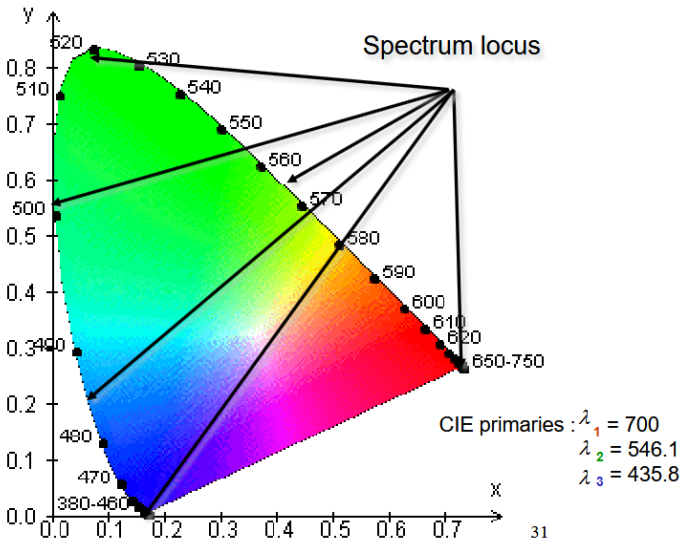



**Figure 47:** CIE color triangle

Inside this triangle, the mix of two colors in some quantities lie on the straight line between those two colors.

Screen standard tries to span as much as this CIE triangle to have the highest fidelity possible. Such color coordinates need 3 primaries on the CIE triangle + the central white point –> 4 points. Also the projection is skewed on the triangle, much more information lie in the bottom blue corner and red corner than in the top green one.

The CIE u-v is a solution which represents area more faithfully.

$$u = \frac{4x}{-2x + 12y + 3} \qquad v = \frac{6y}{-2x + 12y + 3}$$

### 6.1.2. Constancy

Many optical illusions exists where our brain is tricked and interprets colors when there are none or the context can influences our perception. We often interpret the ambient light to infer the result of the dot product on our human cones. This is called the **color constancy**. The brain is trying to infer the color of the object by taking into account the ambient light. This is why we can see a white paper as white even if it is under a red light.

## 6.2. Texture

Many textures exist such as oriented vs isotropic, stochastic vs regular, coarse vs fine, …

### 6.2.1. Fourier features

We analyze the texture based on the power spectrum of the Fourier transform. A peak in the power spectrum indicates a strong periodicity in the image. The orientation of the peak indicates the orientation of the texture. The width of the peak indicates the coarseness of the texture. The more spread out the peak is, the more stochastic the texture is.

One problem with this method is the fact that is captures the global periodicity of the image but not the local one. Moreover, it is not very good at capturing the orientation of the texture. Finally, it is not very good at capturing the coarseness of the texture.

$$\int \int |F(u,v)|^2 dudv$$

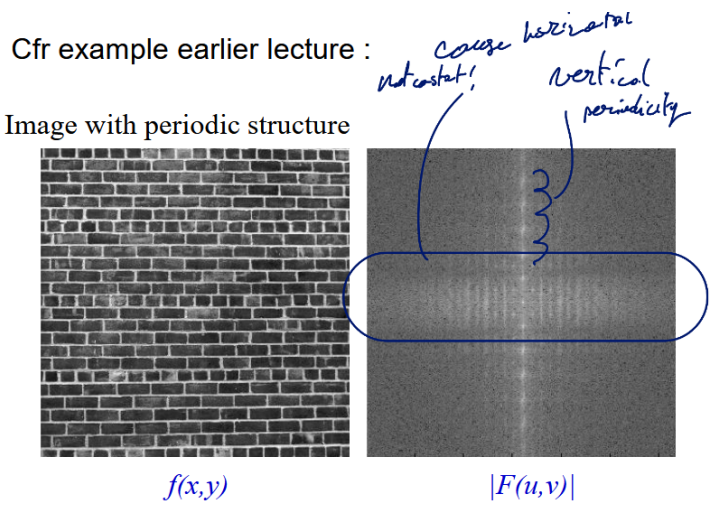



**Figure 48:** Periodicity

### 6.2.2. Cooccurrence matrices

It is inspired by the histogram idea, but this time we will take a chosen offset between two points and count how many i,j pair exists in a double entry matrix. This will give us a measure of the texture based on the co-occurrence of pixel values. We can then derive some statistics from this matrix such as contrast, homogeneity, energy, …

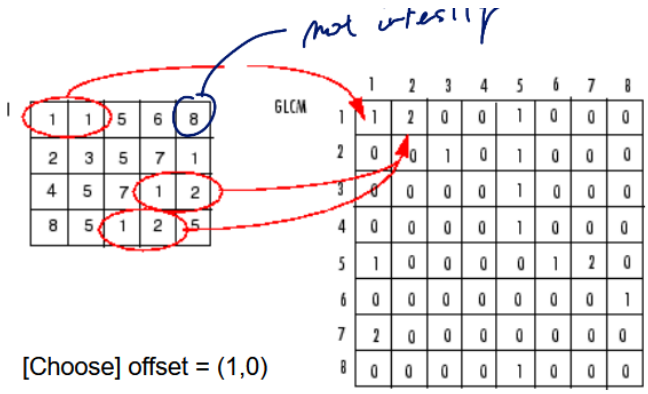



**Figure 49:** Cooccurrence matrix

**Table 6:** Summary table of the features derived from the cooccurrence matrix

| Feature | Expression |
|---|---|
| Contrast | $\sum_{i,j}(i-j)^2 C(i,j)$ |
| Homogeneity | $\sum_{i,j} \frac{1}{1+\|i-j\|} C(i,j)$ |

| Feature | Expression |
|---|---|
| Energy | $\sum_{i,j} C(i,j)^2$ |
| Entropy | $-\sum_{i,j} C(i,j) \log C(i,j)$ |
| Max. Probability | $\max_{i,j} C(i,j)$ |

### 6.2.3. Filter banks

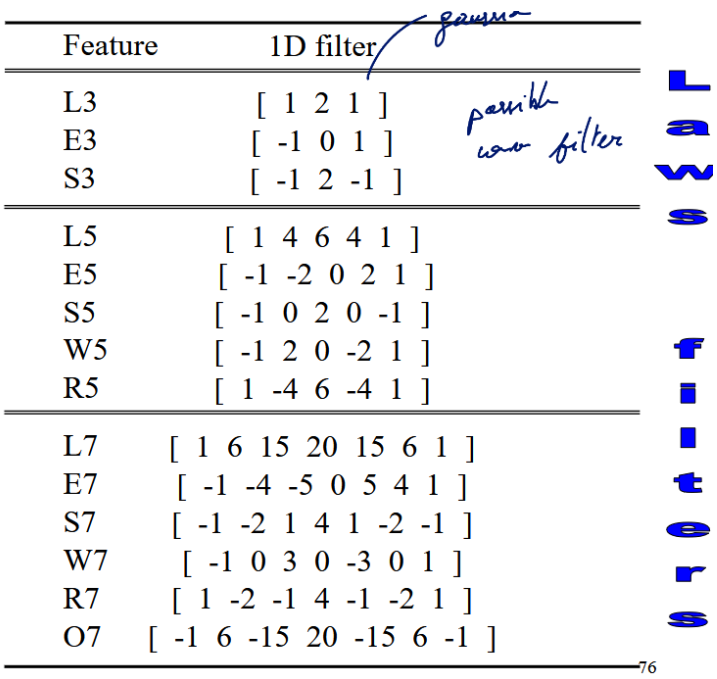



**Figure 50:** Laws filter

**6.2.3.1. Laws** We can start from 1D laws filter to form more advanced decomposed 2D filters. L=Level, E=Edge, S=Spot, W=Wave, R=Ripple. The idea is to have a set of filters that can capture different types of textures. For example, the L5E5 filter will capture edges, while the E5L5 filter will capture spots. We can then apply these filters to the image and analyze the response to get a measure of the texture. It is simle but really effective. Detection or classification nby assigning features / statistical metrics to the outputs.

**6.2.3.2. Gabor** It is a gaussian envelope modulated by a sinusoidal plane wave. It is a band-pass filter that can capture both the frequency and orientation of the texture. The Gabor filter is defined as:

$$g(x,y) = \exp\left(-\frac{x^2+\gamma^2 y^2}{4\Delta_{x,y}^2}\right)\cos\left(2\pi u^* x + \phi\right) \longleftrightarrow G(u,v) = \frac{1}{4\pi\Delta_{u,v}^2}\exp\left(-\frac{(u-u^*)^2+v^2}{4\Delta_{u,v}^2}\right)+\exp\left(-\frac{(u+u^*)^2+v^2}{4\Delta_{u,v}^2}\right)$$

**6.2.3.3. Eigenfilters**

# Part II.

# Modern Image Analysis

# 7. License



## 7.1. Modification

To contribute to this work or any other one from this project please find more information at the Github repository.

## 7.2. Take down

If you are a professor or representing any official party and wish to take down this work, you can submit a takedown request at this url: https://github.com/Tfloow/ESATSummary/issues/new?template=takedown_notice.yml

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